

FBG Sensors for a Smart Glove: A Proof-of-Concept of Contact Force Estimation in a Box-Lifting Task

Juan Pablo Moreno-Garzón¹, Carlos Arturo Narváez Delgado¹, Marcelo Eduardo Vieira Segatto¹,
Carlos A. Cifuentes², *Senior Member, IEEE* and Camilo A. R. Diaz¹, *Member, IEEE*

¹Telecommunications Laboratory (LabTel), Electrical Engineering Department, Federal University of Espírito Santo, Vitória, Brazil

²Bristol Robotics Laboratory, University of the West of England, Bristol, UK

juan.garzon@edu.ufes.br, carlos.delgado@edu.ufes.br, marcelo.segatto@ufes.br, carlos.cifuentes@uwe.ac.uk, camilo.diaz@ufes.br

Abstract—Contact force estimation at the hand has become increasingly important for injury prevention and worker assistance in industrial environments. This work presents a proof-of-concept smart glove with fiber Bragg gratings (FBG) sensors for contact force estimation during box-lifting tasks. Three FBG sensors were integrated into PTFE tubes, embedded in silicone, and attached to a work glove. The sensors were characterized under normal loading up to 10 N and, subsequently, the smart glove was evaluated in repetitive box-lifting trials with three users, using baseline correction, a contact heat-map analysis, and zero-drift assessment. The results obtained with linear regression models showed a predominantly linear behavior, with RMSE values up to 0.47 N during loading (up) and 0.23 N during unloading (down), and hysteresis values in the range of 9.06–19.17%. These results show that contact force estimation using FBG sensors is feasible and support their use in assistance and load-monitoring applications in repetitive tasks.

Index Terms—Fiber Bragg Gratings (FBG), Smart Glove, Box-lifting, Contact Force Estimation, Optical Fiber Sensors.

I. INTRODUCTION

Monitoring hand strength has become increasingly important for improving worker safety and preventing tendon or muscle injuries, especially in repetitive tasks where a conventional glove is not sufficient to protect them [1]. Hand force and pressure measurements are relevant in robotics, rehabilitation, and occupational therapy, as they provide information for diagnosis, prevention, and assistance. Therefore, glove-based measurement systems have gained popularity, generally integrating ultrasonic, magnetic, or optical sensors, among which tactile sensors stand out for being flexible, thin, and lightweight, making them very suitable for monitoring force in activities of daily living (ADL) [2].

Different technologies have been used in this type of system, where flexible and deformation sensors are limited due to their durability under repeated bending or deformation [3]. Another current limitation is the kinematic location of the sensors on the joints where the measurement is to be taken [4]. In addition, other sensors present stability and linear variation issues [5], as well as possible drawbacks arising from interference or sizes that are not suitable for portable applications. Recently, technological advances in materials science and optical fibers have promoted the use of fiber optic sensors as an alternative to electronic sensors, due to their small size, high precision, and strong immunity to interference. Among them, fiber Bragg gratings (FBG) sensors are widely

used for measuring physiological parameters and have been extensively explored in wearable devices [6].

In recent years, applications have emerged in which FBG sensors are integrated into gloves to detect contact, force magnitude, or finger tracking. In one such case, an FBG sensor is used for rotation and contact detection, with data validated by a micro-interrogator and transmitted wirelessly in real time [7]. Another application presented in [8] addresses haptic force detection, where the variation in force applied to the finger when holding different objects was measured. In sensorized gloves that use FBGs, it has been observed that most systems are designed to measure finger joint angles, and very few are designed to measure force or contact. Among existing systems, measurements are usually limited to the distal phalanx. Additionally, commonly used materials include thermoplastic polyurethane (TPU) or teflon tubes to coat the fiber, as well as silicone, which allows the fiber to withstand deformation without breaking [3]. However, to the best of the authors' knowledge, few studies have investigated the use of FBG-based gloves to estimate forces in industrial tasks, such as lifting boxes.

FBG sensors are portable and lightweight and highly sensitive to temperature and deformation, making them suitable for biomedical applications where minimal variations need to be measured. Furthermore, these sensors are not affected by electromagnetic or radio-frequency interference, unlike conventional electrical sensors [9], [10]. The main disadvantage of these systems lies in the cost of the interrogators responsible for demodulating the sensor signal [11]. FBGs are passive optical devices that function as narrowband filters. When an external variation is applied, a shift occurs in their reflection spectrum. By demodulating this information, the changes caused by these external perturbations can be obtained, allowing the measurement of different parameters such as temperature, deformation, and others [11], [12].

In this proof-of-concept study, a smart glove with FBGs is designed to estimate contact forces in box lifting tasks, as a first step toward its integration into an upper limb-exoskeleton, such as the one presented in [13] and other assistive applications. The characterization of three sensors strategically located in the glove is presented, as well as a sensitivity comparison with an unencapsulated sensor. Finally, preliminary tests of a repetitive task are carried out with three

users (U1, U2 and U3), consisting of lifting a box ten times.

II. METHODOLOGY

A. FBG Sensing Principle

FBGs are sensors that reflect a specific range of wavelengths of modes and transmit everything else. This works thanks to the Bragg reflector, which is a periodic variation in the refractive index of the fiber core, which also functions as a dielectric mirror for wavelength selection [14]. The grating of these sensors selectively reflects a narrow band of light that matches the modulation phase of the refractive index of the central layer of this grating, with the central wavelength of the sensor satisfying $\lambda_B = 2n_{\text{eff}}\Lambda$, where λ_B is the central wavelength of the FBG; n_{eff} is the effective refractive index of the fiber core; Λ is the grating period. Factors such as temperature and stress cause the central wavelength of the FBG to change, as presented by (1):

$$\frac{\Delta\lambda_B}{\lambda_B} = (1 - p_e) \Delta\varepsilon + (\alpha + \xi) \Delta T, \quad (1)$$

where p_e is the photoelastic coefficient, α and ξ accounts for thermal effects, and $\Delta\varepsilon$ and ΔT are the strain and temperature changes [15].

B. Sensors Design

Four sensors were manufactured using an ultraviolet (UV) laser (CNI DPS-266, China) operating at a wavelength of 266 nm, emitting 8-nanosecond pulses with a repetition rate of 100 Hz. This required a boron-doped photosensitive fiber (FiberCore, UK), whose core must be exposed to ensure efficient UV irradiation. A phase mask (Ibsen Photonics, Denmark) was used to inscribed the periodic modulation of the refractive index along the fiber core [16]. This allowed the FBGs to be centered at their corresponding wavelengths $\lambda_{B,2} = 1533.99$ nm, $\lambda_{B,3} = 1519.89$ nm and $\lambda_{B,4} = 1525.42$ nm. After obtaining the inscribed fiber, its mechanical strength was improved by embedding it in silicone (Polisil, Brazil). This encapsulation reinforced the fiber and enhanced the sensors' response to axial deformation. To further protect and support the fiber, a polytetrafluoroethylene (PTFE) tube with holes was used, which was also embedded in the silicone and placed in the most sensitive region of the sensor, as shown in Figure 1.

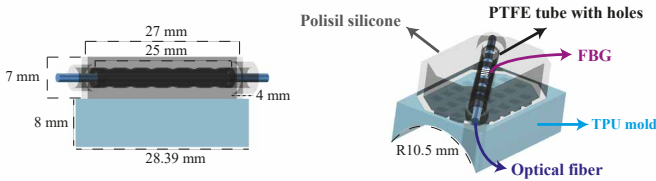


Fig. 1. FBG Sensors design.

One of the sensors was fabricated without the PTFE tube to enable a later sensitivity comparison. The remaining sections of the fiber were covered with 3 mm silicone tubes. To achieve

the sensor structure, a hexagonal mold was designed in TPU, which allowed for more efficient placement of the sensor in the glove. The sensors were then placed at critical grip points of a standard work glove [17], [18] arranged diagonally to allow the optical fiber to pass through the fingers.

C. Experimental Setups

1) *Sensor Characterization Setup*: After manufacturing the sensors and removing the walls of the hexagonal mold, they were mounted on a 3D-printed pressure base. This is done in order to apply normal forces to the sensors to characterize them in an experimental setup. The setup consisted of a micrometer screw (263MXL, Starrett, USA) and a load cell (LCM201, Omega, USA) to apply force to the sensors. As illustrated in Figure 2, this micrometric screw was used to apply a controlled force to the sensor. This force magnitude corresponds to the normal force with respect to the sensor surface. Data collection from the FBG sensors was performed using a HYPERION si255 interrogator (LUNA, Inc., United States) with a sampling frequency of 5 kHz, controlled via ENLIGHT software (Micron Optics, Inc., United States).

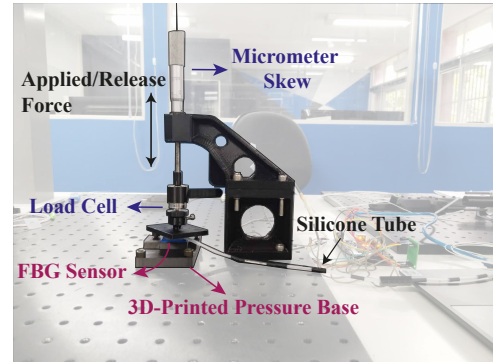


Fig. 2. Experimental setup used for the characterization of the FBG sensors.

2) *Box-Lifting Task Setup*: After characterizing the sensors, they were placed on the distal phalanges of the fingers using a standard work glove, as shown in Figure 3a.

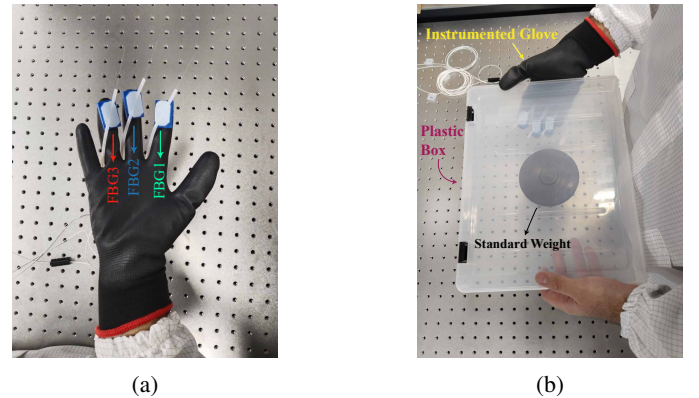


Fig. 3. FBG-based smart glove used in this study: (a) FBGs attached on the fingers, and (b) manual box-lifting task.

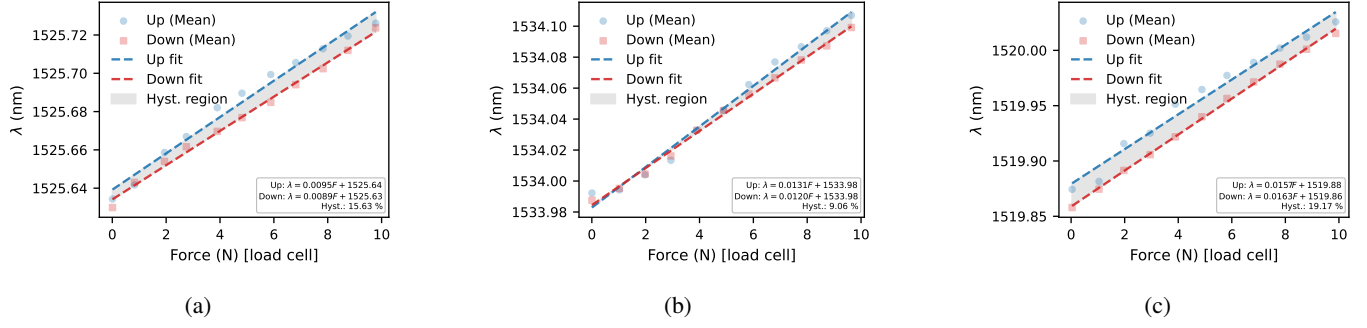


Fig. 4. Hysteresis loops obtained during the static characterization of the three FBG sensors placed on the index (FBG1), middle (FBG2) and ring (FBG3) fingers.

A plastic box weighing 162.53 g with dimensions 34 cm x 26 cm x 4 cm and a standard weight of 999.25 g placed inside the box were used to perform a preliminary test of the sensors. To measure the FBGs-based sensors response, the same interrogator used for their characterization was employed. Furthermore, contact force estimation was carried out using the average sensitivity with the linear equation obtained for each sensor.

D. Procedures

1) *Force Characterization Protocol*: To characterize the sensors, normal forces were applied to each one on the printed base using the experimental setup, as shown in Figure 2. Force levels from 0 to 10 N were applied in increasing and decreasing steps of 1 N, each level being maintained for 15 s, thus obtaining the hysteresis response of the sensor and the sensitivity of each one for contact force estimation. This loading sequence was repeated three times per sensor in order to average the response.

These data were processed per channel using zero-phase Butterworth low-pass filtering, previous-sample outlier replacement, and moving-average smoothing (no downsampling); the same pipeline was applied to the box-lifting trials. The fit was assessed using R^2 , RMSE, and hysteresis. RMSE was computed from the wavelength residuals and converted to force using the fit sensitivity; hysteresis was computed as the loading-unloading difference.

2) *Box-Lifting Task Protocol*: For the preliminary test, the box was lifted for a total duration of 10 minutes and 30s, as shown in Figure 3b. Ten lifting cycles were performed, in each cycle, the box was lifted for 30 s and then released for another 30 s, starting without lifting the box, which is a test similar to the one performed in [19], making it clear that in the hand wearing the glove, only the three sensors make contact with the box, avoiding any other support other than these. Subsequently, the average linear sensitivity equation for each sensor was used to estimate the exerted contact forces. In addition, a baseline of 5–15 s was defined in which the box was not lifted, and the average wavelength value for that interval was calculated. This value was taken as a zero reference and subtracted from the signal before applying the sensitivity, thus ensuring that the

resting value was 0 N. This baseline procedure was adopted to mitigate thermal drift and small preloads introduced when placing the hand on the box in a way that differs from a normal force. Finally, a contact analysis of the sensors with the box was performed using a heat map of the measurements from each sensor in this test, and a zero-drift analysis was carried out throughout the experiment.

III. RESULTS

A. Sensors Characterization

1) *Wavelength–Force Response and Sensitivity*: Upon characterizing each sensor as described in the methodology section, a linear fit was obtained in which the slope indicates the sensitivity of the sensor. In this case, separate linear fits were computed for the up and down data. Table I shows the metrics obtained from these fits for each sensor. As can be seen, the three FBG sensors designed and encapsulated with a PTFE tube obtained an $R^2 > 0.97$, and the FBG2 exhibits the lowest hysteresis, with a value of 9.06%. The hysteresis of each sensor is also analyzed, as can be seen in Figure 4, which presents the linear regressions for the loading and unloading branches of each FBG.

TABLE I. Linear fit metrics for loading (up) and unloading (down) branches and hysteresis for each FBG sensor.

Sensor	R^2		RMSE [N]		Hysteresis [%]
	Up	Down	Up	Down	
FBG1	0.97	0.98	0.41	0.22	15.63
FBG2	0.99	0.99	0.33	0.17	9.06
FBG3	0.98	0.99	0.46	0.12	19.17

2) *Comparison Between Sensor Configurations*: Two sensor designs were compared, the encapsulated configuration with a PTFE tube and a non-encapsulated design in which the FBG was embedded directly in silicone. The non-encapsulated sensor yielded an upward linear regression of $\lambda = 24.70 \text{ pm/N} + 1525.42 \text{ nm}$ ($R^2 = 0.95$) with an RMSE of 0.0174 N, as well as a downward linear regression of $\lambda = 24.30 \text{ pm/N} + 1525.40 \text{ nm}$ ($R^2 = 0.93$) with an RMSE of 0.02 N and a hysteresis of 14.42%. This difference is more clearly illustrated in Figure 5, which shows the average sensitivity lines for each sensor. As can be seen,

the encapsulated design leads to a reduction sensitivity but provides mechanical robustness.

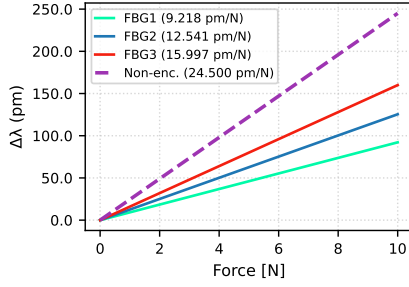


Fig. 5. Comparison of sensitivity of encapsulated and non-encapsulated FBG sensors.

B. Box-Lifting Experiments

1) *FBG Signal Behavior During Lifting Task*: The results of the repetitive box lifting task are presented following the protocol described in the methodology section. Contact force estimation is performed for each sensor, and the lifting intervals are highlighted with a lighter color. The results correspond to the average of the ten lifting cycles performed by the three users, as can be seen in Figure 6. In addition, this figure shows the 5-15 s baseline (gray interval) used for each sensor.

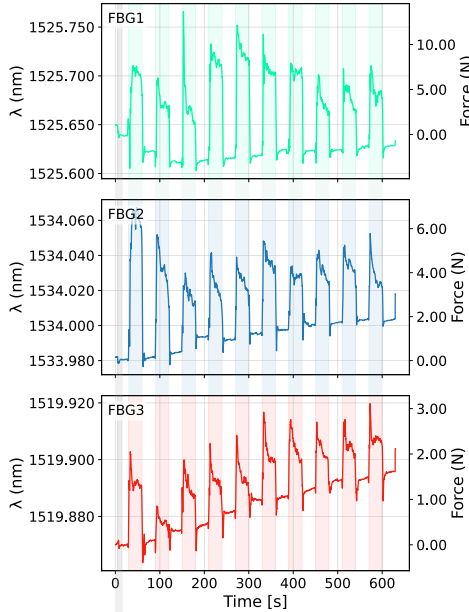


Fig. 6. Box lifting task experiments with the smart glove.

2) *Contact Force Estimation Performance*: Figure 7 presents three heat maps corresponding to the lifts performed by each user, to illustrate the contact exerted on the FBG sensors for each lifting repetition. For each lift, the maximum contact forces measured by each sensor was stored in a sensor-by-lift matrix and then column-wise normalized, so that the

relative contribution of the three sensors added up to the total contact force exerted during that lift, weighting each sensor with a uniform weight of 0.33. This representation makes it possible to observe how the contact is distributed across the detection regions and how it varies between different repetitions.

On the other hand, to analyze the FBGs' behavior at rest during the trials, Figure 8 shows the zero drift of each sensor throughout the experiment. This curve is obtained by extracting and concatenating only the zero-load (rest) segments from the user-averaged lifting signal.

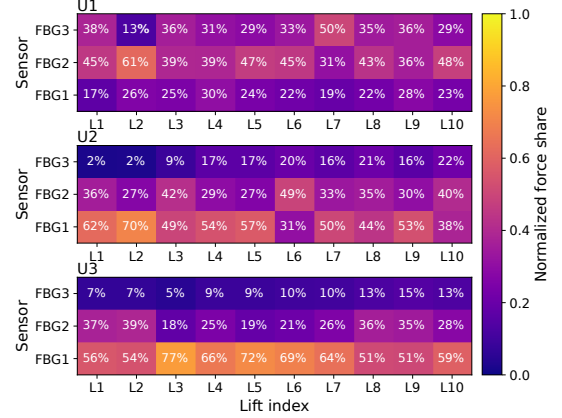


Fig. 7. Contact heat maps for the three FBG sensors across the 10 box-lifting repetitions.

IV. DISCUSSION

The comparison of the sensitivities obtained from the encapsulated FBG sensors and the non-encapsulated sensor was performed, as shown in Figure 5. According to the results, the PTFE-silicone encapsulation and integration into the glove reduce the effective sensitivity to between 40 and 65% of that of the non-encapsulated sensor, since part of the load is absorbed by the flexible PTFE tube and silicone. With this encapsulated design, the sensors gain mechanical robustness despite the loss in sensitivity, which is more appropriate for a repetitive box-lifting task. Among the encapsulated sensors, FBG3 exhibits the highest sensitivity; however, it also shows the largest hysteresis, whereas FBG2 provides an intermediate sensitivity with lower hysteresis. Therefore, the FBG2 appears to offer the best trade-off between sensitivity and repeatability for the current PTFE-silicone design, while the non-encapsulated sensor mainly serves as an upper bound for the achievable sensitivity in the absence of a protective layer and glove-integration constraints, in addition to being more fragile during lifting tasks and more prone to damage.

The results obtained in Table I and Figure 4 show that all sensors exhibited a predominantly linear relationship between wavelength shift and exerted contact force within the established range, with loading and unloading sensitivities ranging from 9.50 to 16 pm/N and coefficients of determination R^2 consistently above 0.98. This indicates that a linear model is

sufficient to describe the quasi-static response of the proposed FBG sensors for the forces considered in this work. In comparison, in [20], a flexible force sensor was used for force measurement with a glove, obtaining R^2 values of 0.94, 0.80, and 0.98 in positions similar to those considered in the present work. In addition, other types of force-sensing gloves can be found in the literature, such as those based on piezoresistive sensors [21].

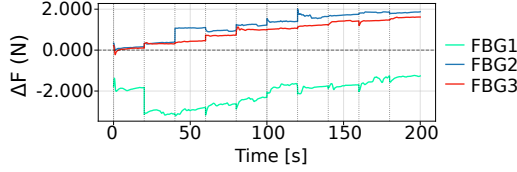


Fig. 8. Zero drift in all three FBG sensors.

On the other hand, the RMSE analysis reveals differences between the loading and unloading branches, as shown in Table I. These results indicate that the response is more repeatable when the force is released than when it is applied, which is consistent with the expected behavior of an encapsulated structure, where friction, slack, and the progressive engagement of the silicone layer can affect the loading path. In any case, all RMSE values remain below 5%, thereby supporting the use of a linear model for quasi-static contact force estimation. The hysteresis values highlight a more pronounced difference between sensors. FBG2 exhibit the lowest hysteresis 9.06%, whereas FBG1 and FBG3 reach 15.63% and 19.17% respectively. From an application standpoint, FBG2 offers the best compromise of linearity, low RMSE, and reduced hysteresis, while the higher hysteresis observed in the other sensors suggests that additional design improvements could further enhance repeatability.

Figure 6 shows the wavelength signals of the three FBG sensors, corresponding to the average of the experiments performed with the three users. A short interval at the beginning of the recording was used to estimate the baseline and remove the offset, compensating for slow drift caused by temperature and small differences in pre-tension between trials. After this correction, a clear pattern appears in each lifting cycle: a rapid increase in wavelength and contact force during the grasp, a quasi-steady plateau while the box is being held, and a decrease when the box is placed back on the support. Pronounced peaks are mainly observed at the beginning of each repetition, which are likely due to dynamic effects such as impacts, small slip events and slight changes in grasp. Since the characterization was performed under controlled normal loading, the FBGs actually measure a combination of normal, shear and bending components that is interpreted here as an equivalent normal force. It has been shown that research on sensing forces other than the normal component, such as shear forces, is still relatively limited, and existing solutions often suffer from directional constraints, reduced range, and fragile or poorly flexible designs [22].

In addition, variations in the contact force levels can be

observed in this test, which are more noticeable in the averaged response of FBG1. These variations arise from the initial contact with the box and from the fact that, as the box is lifted, the grip force increases in proportion to the weight of the box and the friction with the object, leading to higher forces than those expected from simply holding a static load. Moreover, when holding an object, the nervous system continuously adjusts the grip force to prevent slip [23]. The larger variability observed in FBG1 compared to the other sensors may also be related to human error during the fabrication of this sensor, since improper silicone filling inside the PTFE tube cavities can leave voids that induce microvibrations, adding noise. However, the sensor response is acceptable because it remains consistent across the trials with different users.

Figure 7 summarizes the relative contribution of each sensor to the total estimated contact force over the ten lifting repetitions for each user, revealing a user-dependent pattern. For U1, FBG2 carries the largest share of the load (between 40 and 60%), while FBG3 provides a secondary contribution and FBG1 remains below 30% in all lifts. In contrast, for U2 and U3 the dominant contribution comes from FBG1, which accounts for approximately 50–70% of the total contact force, with FBG2 playing a secondary role and FBG3 contributing only marginally. For each user, the relative contributions remain fairly consistent, with variations of about 10–20 percentage points that can be attributed to changes in grasp or hand posture. These results confirm that the distribution of contact is strongly influenced by the user’s grasp strategy and that some sensors provide limited information for contact force estimation in certain users.

These contributions are also affected by the zero drift shown in Figure 8. Although all FBGs start near zero, their estimated force gradually drifts as the repetitions progress. FBG2 and FBG3 exhibit the largest positive drift, reaching nearly +2 N by the end of the experiment, whereas FBG1 shows a comparable negative drift relative to the baseline. This suggests that loading induces residual strain in the instrumented glove. Local boundary conditions, such as bending and shear, may also contribute to the zero offset, and temperature drift is another important factor to consider for this type of sensor [24]. This drift could be addressed by leveraging the trend observed in the FBG sensors during the rest intervals, with the aim of increasing the robustness of the smart glove’s overall response.

A positive correlation would be expected between the contact level and the zero offset; that is, sensors subjected to more loading cycles should exhibit larger baseline shifts due to mechanical creep of the PTFE–silicone encapsulation and temperature effects. However, this relationship appears weak when comparing the drift curves with the contact heat maps. Therefore, the contact heat maps are mainly useful for identifying which sensors are most relevant for contact force estimation (FBG2 and FBG3 for U2, and FBG1 for U1), whereas the zero offset seems to depend more on the mechanics of the encapsulation material than on the mean contact level alone.

V. CONCLUSIONS AND FUTURE WORK

This study demonstrates that it is possible to estimate contact forces in repetitive tasks such as box lifting using FBG sensors integrated into a work glove. The obtained results show that the sensors provide good performance and that, depending on their location on the glove, the user's grasp pattern and a consistent sensor design, some positions achieve better results under the proposed characterization. The analysis of multiple repetitions indicates that, due to the encapsulation material, temperature effects and the direction in which the sensor makes contact with the box, a shift of the initial position or zero drift is induced. The results also reveals which sensors contribute most during the task, showing that the relative contribution of each FBG can vary with the user's grasp strategy. Based on these findings, the FBG-based smart glove emerges as a candidate for assistance and load-monitoring applications in repetitive tasks performed by industrial workers, enabling contact force estimation and providing quantitative feedback on the physical demands of this type of task.

For future work, a multi-axis (including shear) and temperature characterization is planned in order to mitigate drift effects and reduce reading errors caused by contact forces that are not purely normal. In addition, tests with a larger number of participants will be conducted to more thoroughly assess the variability and repeatability of the measurement system. To this end, it may be necessary to include additional sensors to improve coverage and obtain a more robust global contact force estimate from all FBGs, leveraging data-driven approaches such as machine learning and deep learning. A further line of research will focus on integrating the proposed smart glove with industrial upper-limb exoskeletons.

ACKNOWLEDGMENT

This work is partially supported by FAPES (EXTRATO ATA DA 6ª REUNIAO CCAF/2022 –“CPID 2.2”, 1336/2025, 1455/2025), CNPq (306323/2024-9, 404111/2023-8, and 402739/2024-8), and IEEE SPARX. The first and second authors were supported in part by FAPES (2025-2QNJ8) and by CAPES (001), respectively.

REFERENCES

- [1] A. Raza, U. Emanuele, A. Torchia, S. Sinopoli, and A. Cassinese, "Ergonomic smart gloves made on textile for tactile force sensing," *IEEE Sensors Journal*, vol. 25, pp. 29 347–29 354, 2025.
- [2] S. Borik, A. Kmecova, M. Gasova, and M. Gaso, "Smart glove to measure a grip force of the workers," in *2019 42nd International Conference on Telecommunications and Signal Processing (TSP)*. IEEE, 7 2019, pp. 383–388.
- [3] X. Hu, Y. Xu, H. Zhang, J. Xie, D. Niu, Z. Zhao, and X. Qu, "The fiber bragg grating (fbg) sensing glove: A review," *IEEE Sensors Journal*, vol. 23, pp. 11 374–11 382, 6 2023.
- [4] J. Sun, Y. Shen, and J. Rosen, "Sensor reduction, estimation, and control of an upper-limb exoskeleton," *IEEE Robotics and Automation Letters*, vol. 6, pp. 1012–1019, 4 2021.
- [5] P. Xu, D. Xia, J. Li, J. Zhou, and L. Xie, "Execution and perception of upper limb exoskeleton for stroke patients: a systematic review," pp. 557–578, 9 2022.
- [6] D. L. Presti, C. Massaroni, C. S. J. Leitao, M. D. F. Domingues, M. Sypabekova, D. Barrera, I. Floris, L. Massari, C. M. Oddo, S. Sales, I. I. Iordachita, D. Tosi, and E. Schena, "Fiber bragg gratings for medical applications and future challenges: A review," *IEEE Access*, vol. 8, pp. 156 863–156 888, 2020.
- [7] Y. Jiang, V. Reimer, T. Schossig, M. Angelmahr, and W. Schade, "Fiber optical multifunctional human-machine interface for motion capture, temperature, and contact force monitoring," *Optics and Lasers in Engineering*, vol. 128, 5 2020.
- [8] S. Ambastha, S. Umesh, and S. Asokan, "Force sensing for object grasp with fiber bragg grating based wearable haptic device," in *2019 1st International Conference on Advanced Technologies in Intelligent Control, Environment, Computing & Communication Engineering (ICATIECE)*. IEEE, 2019, pp. 280–283.
- [9] S. M. Nizar and B. E. Caroline, "Comparison of fiber optic sensors based on fbg-a review," in *2019 IEEE International Conference on System, Computation, Automation and Networking (ICSCAN)*, 3 2019, pp. 1–6.
- [10] P. Lorenzutti, J. A. Coimbra, R. Menegardo, L. Mayag, E. Souza, E. Morais, A. Frizzera, C. Diaz, and A. Leal-Junior, "Fiber bragg gratings sensor system for kinematic and dynamic assessment in lower limb robotic rehabilitation," *IEEE Sensors Journal*, vol. 25, pp. 19 275–19 281, 2025.
- [11] X. Song, Y. Fan, and X. Tang, "Fbg-based wearable sensors and devices in the healthcare field: A review," *Optics and Laser Technology*, vol. 128, 2 2025.
- [12] C. Lyu, P. Li, J. Zhang, and Y. Du, "Fiber optic sensors in tactile sensing: A review," *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1–16, 1 2025.
- [13] L. J. Arciniegas-Mayag, A. R. Das, D. Casas-Bocanegra, S. Otalora, M. F. Jimenez, M. E. Segatto, C. A. Diaz, M. Munera, and C. A. Cifuentes, "Cable-driven exosuit to assist affected upper-limb users with hemiparesis," in *Proceedings of the IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics*. IEEE Computer Society, 2024, pp. 1629–1634.
- [14] M. Elsherif, A. E. Salih, M. G. Muñoz, F. Alam, B. AlQattan, D. S. Antonyamy, M. F. Zaki, A. K. Yetisen, S. Park, T. D. Wilkinson, and H. Butt, "Optical fiber sensors: Working principle, applications, and limitations," *Advanced Photonics Research*, vol. 3, 11 2022.
- [15] Z. Luo, X. Zhou, and L. Yang, "Temperature compensation study of fbg sensors for strain monitoring in large science devices," in *Proceedings of 2023 IEEE 5th International Conference on Civil Aviation Safety and Information Technology, ICCASIT 2023*. Institute of Electrical and Electronics Engineers Inc., 2023, pp. 343–347.
- [16] J. He, B. Xu, X. Xu, C. Liao, and Y. Wang, "Review of femtosecond-laser-inscribed fiber bragg gratings: Fabrication technologies and sensing applications," pp. 203–226, 6 2021.
- [17] J. Cepriá-bernal and A. Pérez-gonzález, "Dataset of tactile signatures of the human right hand in twenty-one activities of daily living using a high spatial resolution pressure sensor," *Sensors*, vol. 21, 4 2021.
- [18] P.-D. Sarra, V. Fiska, K. Mitsopoulos, D. Mylopoulou, and P. D. Bamidis, "Kinematic and dynamic analysis of the human hand's articulation for wearable soft-robotic device applications," in *10th Panhellenic Conference on Biomedical Technology*. Global Clinical Engineering Journal, 10 2024, pp. 84–88.
- [19] G. Zhou, M. L. Lu, and D. Yu, "Investigating gripping force during lifting tasks using a pressure sensing glove system," *Applied Ergonomics*, vol. 107, 2 2023.
- [20] E. F. Austin, C. P. Kearney, P. J. Chacon, S. A. Wings, P. Acharya, and J. W. Choi, "A fabricated force glove that measures hand forces during activities of daily living," *Sensors*, vol. 22, 2 2022.
- [21] H. L. Horsfall, C. S. Fernandez, B. Bagchi, P. Datta, P. Gupta, C. H. Koh, D. Khan, W. Muirhead, A. Desjardins, M. K. Tiwari, and H. J. Marcus, "A sensorised surgical glove to analyze forces during neurosurgery," *Neurosurgery*, vol. 92, pp. 639–646, 3 2023.
- [22] S. Kerner, M. Krugh, and L. Mears, "Letters wearable shear and normal force sensing glove development for real-time feedback on assembly line processes," in *50th SME North American Manufacturing Research Conference (NAMRC 50, 2022)*, 7 2022, pp. 668–675.
- [23] L. Willemet, F. Roël, D. Abbink, I. Birnieks, and M. Wiertelowski, "Grip force control under sudden change of friction," *Journal of Physiology*, vol. 603, pp. 411–422, 1 2025.
- [24] A. Leal-Junior, M. Silveira, and C. Marques, "A comparison of temperature compensation methods in a diaphragm-embedded fbg," *Results in Optics*, vol. 14, 2 2024.